

Multiple Choice (10 marks)

1. As the number of training examples goes to infinity, your model trained on that data will have:
 - (a) Lower variance
 - (b) Higher variance
 - (c) Same variance
 - (d) None of the above
2. In Yann LeCun's cake, the cherry on top is
 - (a) reinforcement learning
 - (b) self-supervised learning
 - (c) unsupervised learning
 - (d) supervised learning
3. Let us say that we have computed the gradient of our cost function and stored it in a vector g . What is the cost of one gradient descent update given the gradient?
 - (a) $O(D)$
 - (b) $O(N)$
 - (c) $O(ND)$
 - (d) $O(ND^2)$
4. MLE estimates are often undesirable because
 - (a) they are biased
 - (b) they have high variance
 - (c) they are not consistent estimators
 - (d) None of the above
5. Statement 1| RELUs are not monotonic, but sigmoids are monotonic. Statement 2| Neural networks trained with gradient descent with high probability converge to the global optimum.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: Highway networks were introduced after ResNets and utilize max pooling instead of convolutions.
7. True or False: The numerical output of a sigmoid node in a neural network is unbounded, encompassing all real numbers.
8. True or False: Batch Normalization, not Layer Normalization, was utilized in the original ResNet architecture to improve training stability.
9. True or False: Predicting the amount of rainfall based solely on historical data is an unsupervised learning problem.
10. True or False: Dropout randomly sets activation values to one, effectively allowing all neurons to contribute during training.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Critically analyze the statements regarding the ID 3 algorithm's optimality in decision tree generation and the properties of continuous probability distributions. Discuss the implications of each statement.
12. Discuss how you would modify the EM algorithm to compute MAP estimates in a model with latent variables. What specific steps would require changes, and why?

End of Paper